

Part C/MSc

Bayes Methods

Mock 2 2018

Do not turn this page until you are told that you may do so

1. A field is divided into an $m \times n$ grid of square cells. The number $y_{i,j}$ of buttercups in each cell (i, j) is counted. The observation model is

$$y_{i,j} \sim \text{Poisson}(\mu_{i,j}), \quad i = 1, \dots, m, \quad j = 1, \dots, n.$$

Suppose we parameterise $\mu = \phi\psi^T$ as $m \times n$ outer product, where $\phi = (\phi_1, \dots, \phi_m)^T$, $\phi \in [0, \infty)^m$ and $\psi = (\psi_1, \dots, \psi_n)^T$, $\psi \in [0, \infty)^n$ are column vectors of positive parameters.

Thinking of the cells in the matrix μ as a lattice, denote by $N_{i,j}$ the set of labels (a, b) , $a \in 1 : m$, $b \in 1 : n$ for cells adjoining cell (i, j) (interior cells have 4 neighbors, and boundary cells 3, except at the corners where each cell has 2 neighbors). Consider the joint prior density

$$\pi(\phi, \psi) \propto \exp(-\alpha\bar{\phi} - \beta\bar{\psi}) \exp(-\theta \sum_{i=1}^m \sum_{j=1}^n \sum_{(a,b) \in N_{i,j}} |\phi_a \psi_b - \phi_i \psi_j|),$$

where $\bar{\phi} = m^{-1} \sum_{i=1}^m \phi_i$, $\bar{\psi} = n^{-1} \sum_{j=1}^n \psi_j$ and α, β and θ are fixed positive prior hyperparameters.

(a)

- (i) Show that $\pi(\phi, \psi)$ is a proper prior.
- (ii) Write down the posterior distribution $\pi(\phi, \psi|y)$ for ϕ and ψ in terms of the model elements.
- (iii) Write down a simple Metropolis Hastings algorithm targeting $\pi(\phi, \psi|y)$.
- (iv) Let $\chi = (\phi, \psi)$ and $\chi' = (\phi', \psi')$. Suppose the transition density for your algorithm is $K(\chi, \chi')$. Show that K satisfies detailed balance with respect to π . Why is this important?

(b)

- (i) Let $u \sim g(u)$ be a random variable with density g and suppose $(\chi, u) \in \Omega$. Let $f : \Omega \rightarrow \Omega$ be an invertible differential function and suppose that if $(\chi', u') = f(\chi, u)$ then $(\chi, u) = f(\chi', u')$. Given χ we generate an MCMC proposal by simulating $u \sim g(u)$, computing $(\chi', u') = f(\chi, u)$ and taking χ' as our proposal. Write down the acceptance probability for this proposal.
- (ii) Suppose $u_1 \sim U(0.5, 2)$, $u_2 \sim U(0.5, 2)$ and $(\chi', u') = f(\chi, u)$ with $u = (u_1, u_2)$ and

$$\begin{aligned} \phi' &= \bar{\phi} + u_1(\phi - \bar{\phi}) \\ \psi' &= \bar{\psi} + u_2(\psi - \bar{\psi}). \end{aligned}$$

Show that $(\chi, u) = f(\chi', u')$ with $u' = (u'_1, u'_2) = (1/u_1, 1/u_2)$.

- (iii) Show that the acceptance probability is

$$\alpha(\phi', \psi'|\phi, \psi) = \min \left\{ 1, \frac{\exp(-\theta \sum_{i,j} \sum_{(a,b) \in N_{i,j}} |\phi'_a \psi'_b - \phi'_i \psi'_j|) p(y|\phi', \psi')}{\exp(-\theta \sum_{i,j} \sum_{(a,b) \in N_{i,j}} |\phi_a \psi_b - \phi_i \psi_j|) p(y|\phi, \psi)} u_1^{m-3} u_2^{n-3} \right\}$$

where you should define $p(y|\phi, \psi)$, and note any further conditions on ϕ', ψ' .

Hint: Suppose A is an invertible square matrix and u, v are column vectors.

$$\det(A + uv^T) = (1 + v^T A^{-1}u) \det(A).$$

2. (a) Let $S = (S_1, \dots, S_K)$ be a partition of $1 : n = \{1, 2, \dots, n\}$.
- (i) Specify the Chinese Restaurant process for arrivals $1, 2, \dots, n$.
 - (ii) Let $P_{CRP}(S)$ equal the probability the outcome of the CRP is the random partition S . Show that

$$P_{CRP}(S) = \frac{\Gamma(\alpha)\alpha^K}{\Gamma(\alpha+n)} \prod_{k=1}^K \Gamma(n_k)$$

[Note that $\Gamma(\alpha+n) = \Gamma(\alpha) \prod_{i=1}^n (\alpha+i-1)$.]

- (iii) Show that $P_{CRP}(S)$ does not depend on the order of the arrivals.
 - (iv) Let $i_1, i_2, i_3 \in 1 : n$ be three fixed labels. What is the probability that i_1, i_2, i_3 are in the same partition set?
- (b) Consider a mixture of normal densities with a fixed number M of components, Dirichlet distributed mixture component weights $w = (w_1, \dots, w_M)$, and a prior $\pi(\theta^*)$ for the mixture component parameters $\theta^* = (\theta_1^*, \dots, \theta_M^*)$:

$$\begin{aligned} w &\sim \text{Dirichlet}(\alpha_1, \dots, \alpha_M) \quad \text{with } \alpha_M = \alpha/M \text{ for } \alpha > 0 \text{ fixed;} \\ z_i &\sim \text{Multinom}(w), \quad i = 1, \dots, n; \\ \theta_m^* &\sim \pi(\theta_m^*), \quad m = 1, \dots, M. \end{aligned}$$

Here $z_i \sim \text{Multinom}(w)$, $i = 1, \dots, n$ means that $z_i = m, m \in \{1, \dots, M\}$ with probability w_m . In this model $z_i \in \{1, \dots, M\}$ is the label of the cluster to which y_i belongs. The observation model is

$$y_i \sim f(y_i; \theta_{z_i}^*), \quad i = 1, \dots, n.$$

Suppose the list $z = (z_1, \dots, z_n)$ of cluster labels contains $K \leq M$ unique values $m_1, \dots, m_K$. For $k = 1, \dots, K$ let $S_k = \{i : z_i = m_k, i = 1, \dots, n\}$. Let $S = (S_1, \dots, S_K)$. We write $S = S(z)$ for the partition determined from z in this way.

- (i) Write down the posterior for $\theta^*, z, w|y$ in terms of the model elements.
- (ii) Calculate the marginal prior probability $\pi_z(z) = \int \pi_{z|w}(z|w)\pi_W(w)dw$ for a set of cluster labels.
- (iii) For $k = 1, \dots, K$, let $n_k = |S_k|$ give the number of items in cluster k . Let

$$P(S) = \sum_{z: S(z)=S} \pi_z(z)$$

denote the prior distribution over partitions. Show that

$$P(S) = \frac{\Gamma(\alpha)}{\Gamma(\alpha/M)^K} \frac{M!}{(M-K)!} \frac{\prod_{k=1}^K \Gamma(\alpha/M + n_k)}{\Gamma(\alpha+n)}.$$

- (iv) Show that the prior distribution over partitions converges to a CRP as $M \rightarrow \infty$.
- (v) Write down the distribution to which the marginal posterior distribution of $\theta^*, S|y$ converges in the limit as $M \rightarrow \infty$.

3. Let A be the points in the unit square $[0, 1]^2$. For $i = 1, \dots, n$, let $x_i = (x_{i,1}, x_{i,2})$ be points $x_i \in A$ and let $x = \{x_1, \dots, x_n\}$ be a set of n points. The number and locations of the points are distributed according to a Strauss point process with parameters $\lambda > 0, 0 \leq \gamma \leq 1$ and $R \geq 0$. In particular, let $c(x; R)$ be the function

$$c(x; R) = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \mathbb{I}_{|x_i - x_j| < R}$$

counting the number of R -close pairs of points in x . Let Ω_n be the set of distinguishable point patterns $x = \{x_1, \dots, x_n\}$ in A and let $\Omega = \cup_{n=0}^{\infty} \Omega_n$. The distribution of $x|\lambda, \gamma, R$ is

$$p(x|\lambda, \gamma, R) = \xi(\lambda, \gamma, R) \gamma^{c(x, R)} \lambda^n,$$

where $\xi(\lambda, \gamma, R)$ is a normalising constant ensuring the density is normalised to one over Ω .

(a) [5 marks]

- (i) Give an expression for the normalising constant $\xi(\lambda, \gamma, R)$ for $p(x|\lambda, \gamma, R)$.
- (ii) Suppose $\gamma = 1$. Show, by integrating $p(x|\lambda, \gamma, R)$ over $x \in \Omega_n$ at fixed n , that the number of points n in the random set x has a Poisson distribution.

(b) [8 marks] Suppose x is observed and we wish to estimate λ, γ, R .

- (i) Write down the posterior density $\pi(\lambda, \gamma, R|y)$ giving the density in as much detail as you can but without specifying priors.
- (ii) Suppose an ABC algorithm approximately targeting $\pi(\lambda, \gamma, R|y)$ and based on summary statistics $S(x)$ is available. The ABC algorithm yields samples $(\lambda^{(t)}, \gamma^{(t)}, R^{(t)})$, $t = 1, 2, \dots, T$ with associated statistics $S^{(t)}$. Explain how to make a regression adjustment to the sampled $\gamma^{(t)}$ -values. What is the purpose of this adjustment and how is it justified?
- (iii) What exact distribution does your ABC algorithm target? Briefly interpret the approximation being made in ABC.

(c) Consider a reversible jump algorithm simulating the Strauss process $x \sim p(x|\lambda, \gamma, R)$. The algorithm is structured as follows. Suppose $X_t = x$ with $x = (x_1, \dots, x_n)$ so point pattern x has n points in it.

To add a point pick a new point $z \sim A$ (throw a dart at A). Let $x' = x \cup z$ and $n' = n + 1$. The number of R -close pairs goes up to $c(x')$ and the number of points goes up to $n + 1$.

To delete a point, choose a point $x_i \in x$ at random from $i = 1, 2, \dots, n$. Let $x' = x \setminus x_i$ and $n' = n - 1$. The number of R -close pairs goes down to $c(x')$ and the number of points goes down to $n - 1$.

- (i) Calculate the acceptance probability $\alpha(x'|x)$ when x' differs from x by the addition of one point.
- (ii) Calculate the acceptance probability $\alpha(x'|x)$ when x' differs from x by the deletion of one point.
- (iii) Outline a reversible jump MCMC algorithm targeting $p(x|\lambda, \gamma, R)$.